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A study of the palaeomagnetism of subglacial basalts, SW Iceland: a comparison with oceanic crust

J. Helgason,[1] N. A. Van Wagoner[2] and P. J. C. Ryall[3]

[1]*Science Institute, University of Iceland, Dunhaga* 3, 101 *Reykjavik, Iceland*
[2]*Department of Geology, Acadia University, Wolfville, Nova Scotia, Canada,* B0P 1X0
[3]*Department of Geology, Dalhousie University, Halifax, Nova Scotia, Canada,* B3H 3J5

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SUMMARY

Subglacial volcanic sequences in Iceland were used as analogues of oceanic crust to get an indication of the degree of primary heterogeneity of magnetic directions that can be expected within subaqueous volcanic sequences erupted in both deep and shallow water. In the Icelandic sequences the degree of tilt, brecciation and the amount of hyaloclastite are known so it is possible to test for primary magnetic heterogeneity.

We measured magnetic directions of 119 oriented core samples (2.5 cm diameter) collected from five sites in SW Iceland where there is good exposure in Quaternary subglacial volcanic sequences. Sample sites vary from compact pillowed flows (most analogous to deep water volcanic sequences) to pillow breccia with up to 40 per cent interpillow hyaloclastite (more analogous to shallow water sequences).

Magnetic directions of multiple samples from 16 individual pillows have angular differences of 2°–66° with most being less than 10°. Of the five sites sampled, two sites have inclinations that approximate that of the centred axial dipole, one site has steeper inclinations and two sites have shallower inclinations. When the site averages of magnetic direction with their circles of α_{95} are plotted, there is little overlap. There is little correlation between either the scatter of magnetic measurements or the percentage of non-dipolar inclinations and the occurrence of pillow breccias, hyaloclastite or depositional dips.

We doubt the tectonic significance of small amounts of tilt ($<30°$) estimated from drill cores on the basis of palaeomagnetic inclination as it is not uncommon for rocks to rotate by as much as 30° after they cool through the Curie temperature or for such a scatter to be the result of secular variation. With this in mind, the same caution must be applied when orienting seafloor basalts for palaeomagnetic or structural studies using geopetal features such as drain-away ledges or segregation vesicles, because the rocks may rotate after these features form (at 1200°–1000 °C) but before the rocks cool through the Curie temperature.

Key words: Iceland, palaeomagnetism, subglacial pillows.

INTRODUCTION

The purpose of this study was to get an indication of the degree of 'primary heterogeneity' of magnetic directions that can be expected within subaqueous volcanic sequences which erupted in both deep and shallow water. 'Primary heterogeneity' is used here to refer to variations in magnetic directions caused by rotations as the rock cools through the Curie temperature, but prior to tectonic (or regional structural) events. 'Secondary heterogeneity' refers to variations in magnetic directions related to structural disturbance. This study is important because palaeomagnetic studies of some Deep-Sea Drilling Project (DSDP) cores which penetrated oceanic basement crust revealed variations in the direction of magnetization in a vertical section (e.g. Hall & Robinson 1979). However, debate persists over whether the magnetic variations in a vertical section of oceanic drill core record secular variation in the Earth's magnetic field, secondary magnetization, rotation following magnetization, or episodic volcanism on a slowly spreading

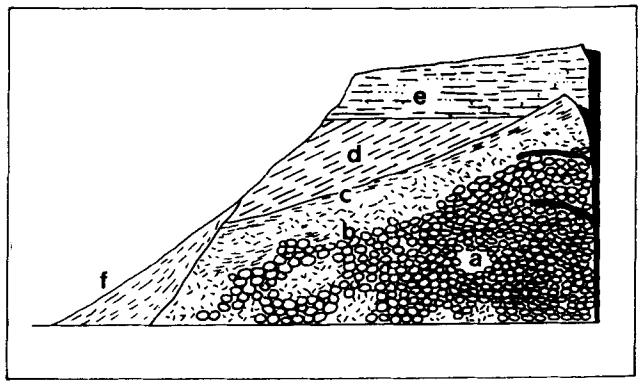


Figure 1. General features of a subglacial ridge in cross-section (from Saemundsson 1967). The main components are: (a) compact pillow lavas, (b) pillow breccia, (c) hyaloclastite, (d) reworked foreset hyaloclastite breccia, (e) subaerially erupted lavas, and (f) scree cover.

ridge (e.g. Hall 1976; Harrison 1976; Harrison & Watkins 1977; Ryall *et al.* 1977; Johnson 1978; Johnson & Merrill 1978; Schouten & Denham 1979; Denham & Schouten 1979; Verosub & Moores 1981; Bideau, Prevot & Lecaille 1984; Van Wagoner 1986).

Subglacial volcanic sequences in Iceland provide an accessible analogue of oceanic crustal layer 2. Some of these sequences were used to investigate the homogeneity of palaeomagnetic directions both within pillows and between pillows of the same formation where the rocks are fresh and the degree of tilt, brecciation and amount of hyaloclastite are known.

The main lithologic components of a subglacial volcanic sequence are (Fig. 1) (a) thick compact pillow flows which form from the first extruding lava if the ice or water load is sufficent; (b) pillow breccia which may form due to a decrease in the overlying pressure as the volcanic section builds up; (c) hyaloclastite which forms as eruptions reach sea level; (d) reworked foreset hyaloclastite breccia; and (e) a subaerial cap of non-pillowed flows (Saemundsson 1967; Allen, Jercinovic & Allen 1982). Both vertical and lateral variations are significant. Following extrusion, the sequence may be subjected to gravitation sliding, collapse, and erosion (Furnes & Fridleifsson 1979).

In SW Iceland, glacial volcanic ridges are commonly about 2 km wide and 20–30 km long, and range in height from <100 to about 500 m above the surrounding plateau. Formation of the sequences of compact pillow flows is considered analogous to extrusion at mid-ocean ridges where the water depth exceeds the volatile fragmentation depth (100–800 m) (e.g. Moore & Friske 1969; Allen 1980; Moore & Schilling 1973; Staudigel & Schmincke 1984; Fischer & Schmincke 1984). The other lithologic types form in shallower water.

In Iceland the thickest ice cover is estimated to have been <1000–1500 m (Walker 1965). Therefore, vesiculation is typically much higher in Icelandic subglacial pillows than in deep submarine lavas. The examples for this study were collected in SW Iceland where the ice cover was <400–600 m (based on the height of subglacially formed strata above the surrounding plateau). The samples used generally contain 5–40 per cent vesicles.

SAMPLING

Samples were collected at five sites (A–E) in SW Iceland where there is good vertical and lateral exposure of Quaternary subglacial volcanic sequences (Fig. 2 and Table 1). Oriented samples were taken from 98 pillows using a two-stroke portable drill. The drill recovers samples 2.5 cm in diameter and several cm long. Within each site several vertical sections were sampled, at 10–40 m intervals. A number of pillows were sampled at each outcrop, here referred to as subsites, with two to three samples collected from some pillows (cores A, B and C). Thus, sample

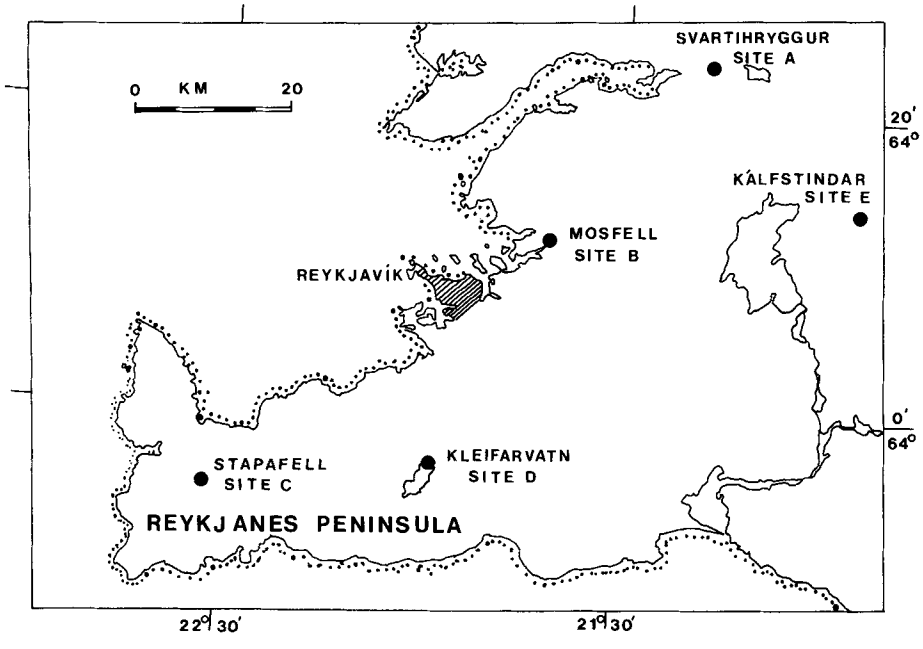


Figure 2. Location of sampling sites A to E in SW Iceland.

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Table 1. Description of subsites.

Subsite	Elevat. (m)	Description	Depositional Dip	Thickness (m)
Site A: Svartihryggyur (64°24'N, 21°18'N)				
A1	140-147	Compact pillowed flow	32°	6-7
A2	148-152	Compact pillowed flow	20°	3-6
A3	157-163	Compact pillowed flow	0°-20°	2.5-10
A4	170-174	Compact pillowed flow	15-25°	3-4
A5	186-191	Ameboid pillow breccia with 20% interpillow hyaloclastite	45°	4-6
A6	216-222	Ameboid pillow breccia with 10% interpillow hyaloclastite	40°	6-8
A7	230-237	Pillowed flow and breccia with up to 40% interpillow hyaloclastite	40°	6-8
A8	250-270	Compact pillowed flow	2°-4°	20
A9	285-290	Sheet flow interbedded with vaguely bedded hyaloclastite	<2°	5
Site B: Leirvogsa River (64°11'N, 21°39'N)				
B1	40	Compact pillowed flow	0°	6-8
B2	40	Large pillow lobe	0°	6-7
B3	40-53	Compact pillowed flow with less than 5% inter-pillow hyaloclastite	0°	13
B4	53-60	Compact pillowed flow	20°	6-7
B5	46-52	Compact pillowed flow	0°	6-8
Site C: Stapafell (63°54'N, 22°31'W)				
C1	58	Compact pillowed flows	0°	35
C2	60	Pillow breccia with 5-10% hyaloclastite	0°	20
Site D: Lake Kleifarvatn (63°56'N, 21°56'W)				
D1	135	Compact pillowed flows	5°	10
Site E: Kalfstindar (64°14'N, 20°52'W)				
E1	191	Lenticular compact pillowed flows intercalated with thin beds of hyaloclastite, 30-40 cm thick. Composite thickness of E1 to E3:190m	0°	-
E2	313	Compact pillowed flows	30°	-
E3	380	Compact pillowed flows	30-40°	-
E4	480	Sheet flows and flow lobes intercalated with hyalo-clastite. On top of sub-site E4 is hyaloclastite, 70 m thick.	<5°	10-15

A8-11C denotes the third core taken from pillow 11 in subsite 8 of site A. Samples were oriented by geographical and sun sighting. Because of concentric bands of vesicles that form in pillows, the cores sometimes break prior to orientation. Consequently we drilled short cores, just long enough for magnetic measurements.

Sample sites vary in physical characteristics from compact pillowed flows to pillow breccia with up to 40 per cent interpillow hyaloclastite (Table 1). Only the more coherent pillowed portions of the breccia were sampled. These portions of the breccias were selectively sampled because they are most closely equivalent to the longer pieces of basalt recovered by ocean drill cores that would be used for palaeomagnetic studies.

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SITE DESCRIPTIONS

Site A: Svartihryggur Ridge

All pillows at site A are aphyric and highly vesicular (*c.* 35 per cent), fresh tholeiitic basalt. Some vesicles are half-filled with subhorizontal layers of opal. Opal also occurs on cooling fractures as 0.5–1 mm thick horizontal and subparallel veins. The lower part of the formation comprises lenticular, compact pillowed flows which exhibit foreset bedding with depositional dips up to 45°. The lenses are 2–10 m thick, and up to 14 m long (down dip). The sequence grades upward to a pillow breccia comprising isolated pillows in a hyaloclastite matrix, overlain by hyaloclastite with pillow block breccia of large but irregularly shaped pillows. Fine-grained, reworked hyalocl-

Table 2. Palaeomagnetic results. (a) For pillows. Sample designation e.g. A2–3A. A = site, 2 = subsite, 3 = pillow, A = core. (b) For subsites.

(a)

SAMPLE	J_o	MDF	I	D	J_o (σ)	I (σ)	I	D (α95)
A2-3A	9.76	460	68.1	356.5				
A2-3B	11.17	440	76.9	21.4	10.27 (0.78)	74.5 (5.6)	74.8	8.3 (10.2)
A2-3C	9.89	390	78.5	15.6				
A3-1A	10.18	310	68.7	49.0				
A3-1B	8.62	390	70.1	42.3	9.4 (1.10)	69.4 (1.0)	69.4	45.8 (6.1)
A3-2A	10.63	400	69.1	45.3				
A3-2B	6.27	500	72.8	71.3	9.14 (2.49)	71.2 (1.6)	72.0	51.0 (10.4)
A3-2C	10.53	460	71.8	36.2				
A3-3A	10.12	530	68.4	26.2				
A3-3B	10.65	420	71.7	26.7	10.39 (0.37)	70.0 (2.3)	70.0	26.5 (7.0)
A4-6A	13.33	325	61.6	27.7				
A4-6B	10.01	370	67.5	44.7	11.05 (1.98)	65.8 (3.6)	66.0	38.0 (8.3)
A4-6C	9.81	360	68.2	44.4				
A6-2A	10.49	450	69.9	3.4				
A6-2B	9.84	410	69.9	16.9	9.29 (1.55)	71.7 (3.2)	71.8	10.8 (3.0)
A6-2C	7.54	600	75.4	4.1				
A7-6A	8.31	420	71.3	345.2				
A7-6B	8.81	320	66.4	355.0	8.56 (0.35)	68.8 (3.5)	68.9,	350.6 (13.2)
A8-1A	8.11	290	69.1	356.5				
A8-1B	7.83	370	71.4	356.6	7.92 (0.16)	69.2 (2.2)	69.2	359.0 (4.0)
A8-1C	7.82	360	67.0	3.2				
A8-9A	10.54	410	63.8	337.2				
A8-9B	10.61	390	59.9	357.4	10.58 (0.15)	61.8 (2.8)	62.2	347.9 (22.5)
A8-11A	10.24	340	78.6	42.9				
A8-11C	9.77	360	86.5+	121.5+	10.01 (0.33)	82.6 (6.6)	83.7	58.8 (24.7)
B3-8A	16.97	400	-6.9	112.6				
B3-8B	20.21	390	68.6	329.7	18.59 (2.29)	N/A		
B4-9A	13.69	330	79.7	332.0				
B4-9B	14.43	450	78.4	357.0	14.06 (0.52)	79.0 (1.0)	79.3	345.2(10.6)
B5-8A	12.70	500	81.5	298.0				
B5-8B	13.47	510	72.7	294.3	13.09 (0.54)	77.1 (6.3)	77.1	295.6 (19.3)
C1-23A	17.45	340	77.6	293.7				
C1-23B	14.27	140	77.8	88.5	15.86 (2.25)	77.7 (0.1)	87.3	9.0 (54.9)
C1-31A	21.21	140	84.6	159.4				
C1-31B	17.11	170	81.7	282.7	19.16 (2.90)	83.1 (2.0)	86.5	62.4 (26.7)
C1-32A	14.04	280	81.6+	9.5+				
C1-32B	13.35	190	81.1	5.4	13.70 (0.49)	81.4 (0.4)	81.4	7.4 (1.7)

J_o = Intensity of NRM in units of Am^{-1}.

MDF = Median Demagnetizing Field - The AC Demagnetizing Field (in oersteds) at which J_{NRM} is reduced to half the initial value.

I, D, = Inclination and Declination, averaged over 12 demagnetizing steps except where shown otherwise.

J_o (σ) = Average Intensity of NRM with standard deviation.

I (σ) = Average Inclination with standard deviation (averaged using inclinations only).

I, D (α95) = Average Inclination and Declination with radius of circle of 95% confidence.

+ = Inclination & Declination from fewer than 12 demagnetizing steps.

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Table 2. (*continued*)

(b) PILLOW	J_o	MDF	I	D	SUBSITE J_o (σ)	AVERAGE I (σ)	I	D (α_{95})
A2-1A	9.71	500	74.3	55.4	9.99 (0.40)	74.4 (0.2)	75.5	33.8 (25.1)
A2-3*	10.27	430	74.6	11.6				
A3-1*	9.40	350	69.4	45.8	9.64 (0.66)	70.5 (1.4)	70.8	40.6 (6.8)
A3-2*	9.14	470	72.1	50.4				
A3-3*	10.39	480	70.0	26.5				
A4-6*	11.05	350	66.0	38.0	11.05	66.0	66.0	38.0
A5-1	8.72	340	68.3	0.0	9.27 (1.61)	68.3 (3.1)	68.3	8.4 (4.2)
A5-2	7.61	490	72.5	2.7				
A5-3	11.45	410	65.1	16.7				
A5-4	9.31	490	67.2	7.1				
A6-2*	9.29	490	71.8	8.5	8.83 (0.65)	68.0 (5.4)	67.9	2.0 (15.8)
A6-3	8.37	300	64.1	357.3				
A7-1	10.51	400	75.0	67.8	9.54 (1.38)	71.8 (4.4)	75.5	21.6 (52.2)
A7-6*	8.56	370	68.7	350.7				
A8-1*	7.92	340	69.2	359.0	9.43 (1.43)	69.0 (8.0)	69.9	354.5 (10.1)
A8-4	10.78	340	65.1	5.3				
A8-9*	10.58	400	62.2	347.9				
A8-10A	7.86	420	65.7	344.2				
A8-11*	10.01	350	83.7	58.8				
A9-A	9.17	230	76.1	342.5	7.58 (2.18)	78.3 (2.0)	84.7	326.7 (20.0)
A9-E	5.09	270	78.9	335.7				
A9-H	8.48	390	79.9	177.9				
B1-1	20.20	360	78.7	315.6	18.34 (3.18)	78.1 (0.9)	78.1	351.1 (8.9)
B1-3	20.14	400	77.0+	357.2+				
B1-4	14.67	330	78.5	356.4				
B2-1	10.53	130	72.9	326.8	12.41 (2.65)	72.2 (0.9)	72.2	325.9 (3.5)
B2-2	14.28	130	71.6	325.0				
B3-1	13.41	130	78.2	345.4	14.31 (2.19)	74.9 (3.8)	75.8	335.3 (3.4)
B3-2	9.47	150	71.2	32.9				
B3-3	16.52	320	70.6	345.6				
B3-4	16.50	550	73.6	331.6				
B3-6	13.07	500	82.0	322.3				
B3-7	16.96	490	71.3	349.4				
B3-9	15.77	400	74.6	314.3				
B3-10	12.64	420	80.5	313.0				
B3-11	14.52	350	73.8	325.9				
B3-12	14.66	380	75.2	347.2				
B3-13	12.49	280	74.2	326.5				
B3-14	15.71	240	-29.2	251.6				
B4-1	11.71	450	75.4	347.4	12.36 (3.07)	75.2 (2.1)	75.7	353.8 (2.4)
B4-2	11.56	480	75.3	354.0				
B4-3	11.87	490	76.7	358.4				
B4-4	16.72	420	74.0	0.9				
B4-5	13.82	420	73.2+	354.8+				
B4-9*	14.06	390	79.0	345.3				
B4-13	6.77	500	73.4+	82.4+				
B5-7	12.61	550	77.3+	295.4+	13.21 (0.46)	79.1 (2.4)	79.3	297.7 (2.9)
B5-8*	13.09	500	77.1	295.6				
B5-9	13.57	500	81.8	296.8				
B5-10	13.56	510	80.9	304.7				
C1-6	14.77	310	82.7	301.4	16.71 (2.77)	82.9 (4.1)	85.2	338.0 (6.8)
C1-15	20.06	200	77.0	308.3				
C1-23*	15.86	240	87.3	91.1				
C1-31*	19.16	160	86.2	64.3				
C1-32*	13.70	240	81.3	8.8				
C2-1	13.81	600	82.4	330.7	17.23 (2.15)	82.9 (4.4)	87.1	273.5 (5.7)
C2-2	17.13	350	88.7	326.6				
C2-3	20.71	460	83.7	175.7				
C2-4	17.67	460	87.4	296.4				
C2-5	16.93	220	76.0	287.9				
C2-8	19.38	410	85.8	75.5				
C2-9	16.86	400	78.3	154.5				
C2-10	15.32	440	80.6	295.3				

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Table 2. (*continued*)

D1-2	16.44	190	67.9	349.9	15.91 (4.40)	67.7 (7.8)	71.8	335.7 (8.3)
D1-3	10.96	200	71.4	345.4				
D1-8	14.48	120	53.4	343.2				
D1-12	12.33	100	78.1+	343.7+				
D1-13	15.43	120	76.4	287.8				
D1-16	26.33	180	74.6	173.0				
D1-20	11.38	170	71.7	332.4				
D1-21	20.60	400	69.7	343.9				
D1-27	12.10	130	61.5	285.6				
D1-29	16.36	140	65.2	332.6				
D1-30	16.04	200	65.8	351.8				
D1-34	18.52	220	58.3	3.2				
E1-1	9.03	>500	69.5	352.4	10.32 (1.04)	69.4 (3.3)	69.3	353.4 (2.6)
E1-2	10.54	500	67.2	354.2				
E1-3	10.05	540	71.6	354.9				
E1-4	10.04	500	73.6	354.8				
E1-5	12.18	400	64.5	351.4				
E1-6	10.05	450	69.5	356.2				
E2-1	9.48	510	63.9	1.2	10.91 (1.58)	64.8 (1.2)	69.9	3.0 (2.1)
E2-2	12.90	390	65.9	359.2				
E2-4	9.82	540	66.0	5.1				
E2-5	11.43	400	63.8	6.2				
E3-1	7.44	510	69.1	5.6	15.04 (11.71)	67.4 (1.8)	67.3	3.9 (2.9)
E3-3	28.52	160	65.6	2.3				
E3-4	9.15	80	67.2	3.9				
E4-1	9.02	110	66.2	209.0	9.94 (3.98)	68.3 (6.8)	68.8	2.2 (5.7)
E4-2	4.31	100	8.3	123.1				
E4-3	8.99	100	64.8	3.6				
E4-6	6.55	140	68.5	11.6				
E4-7	11.05	360	64.9	357.2				
E4-9	11.41	500	68.2	355.4				
E4-10	8.51	580	78.5	354.2				
E4-11	11.04	350	58.1	6.8				
E4-12	18.61	120	77.6	1.7				

Heading symbols as Table 2a.
Δ = not used in averaging.
+ = inclination and declination from fewer that 12 demagnetizing steps.
* = pillow average from Table 2a.

astite caps these units. Based on the dip of lavas that bank up against the eastern side of Svartrihryggur it is estimated that this formation was tilted by a maximum of 5° to the east, toward the neovolcanic zone.

This formation is overlain by subaerially erupted lavas of reverse polarity, R2 of Einarsson (1957) and Sigurgeirsson (1957). We correlate the Svartihryggur sequence of normal magnetic polarity with subchron N3 (Einarsson 1957, 1962; Sigurgeirsson 1957) which was tentatively identified as the normal Reunion event (Kristjansson, Fridleifsson & Watkins 1980).

Nine subsites were sampled at Site A (Table 1). Magnetizations were determined for 36 samples (Table 2a and b). These samples were collected over a vertical distance of 140 m.

Site B: *Leirvogsa River*

The rocks at Leirvogsa River are olivine tholeiites (Furnes & Fridleifsson 1978). The pillows that were sampled are fresh, aphyric and sparsely vesicular.

The degree of post-depositional tilting of a maximum of 2°–4° to the east or southeast was determined by the attitude of drain-away ledges in pillow lobes; these ledges are geopetal features, diagnostic of the original cooling position of the rocks (Ballard *et al.* 1975; Prevot *et al.* 1976; Wells, Bryan & Pearce 1979; Bideau *et al.* 1984).

This section was correlated with the Mosfell formation located 500 m to the east of Site B (Furnes & Fridleifsson 1979; Furnes *et al.* 1980; Fridleifsson, Furnes & Atkins 1982), which is less than 0.5 Ma old (Furnes & Fridleifsson 1978). The normal magnetic signature at site B is, therefore, correlated with the Brunhes polarity epoch.

Five subsites were sampled at Leirvogsa (Table 1). Magnetizations were determined on 32 samples (Table 2a and b). These sites span a vertical distance of only 20 m, but were collected over a distance of 185 m along strike.

Site C: *Stapafell*

Rocks at Stapafell are olivine tholeiites (e.g. Yagi 1964; Mathews, Thorarinsson & Church 1964). Stapafell is a subglacially formed pillow lava ridge located within the volcanically active Reykjanes fissure swarm which extends into sea 10 km further to the southwest (Jakobsson, Jonsson & Shido 1978). At the NW side of Stapafell the pillow lava flowed into a small graben formed by the Reykjanes fissure swarm. The horizontal orientation of most drain-away ledges in pillow lobes indicates that no post-deposition tilting has occurred.

On the basis of its young morphology we believe the Stapafell formation formed during the last glacial (Weischel).

Two subsites were sampled across two traverses on the NW and SW sides of the ridge about 400 m apart (Table 1). On the NW side of Stapafell ridge, the sequence comprises a

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compact sequence of pillowed flows. Drain-away ledges commonly occur in pillows greater than 1 m diameter. These ledges are mostly horizontal but occasionally dip up to 20° in random directions. Because of the random orientations of these dipping ledges, it is probable that tilting took place prior to complete solidification of the pillows, either as a result of differential settling of the cooling pillowed flow, or as a result of compaction of these pillows due to burial under lava from a new effusive phase. This is an important consideration when trying to use such ledges to orient seafloor basalt samples collected by dredge and submersible for palaeomagnetic studies. The rocks of the traverse on the southwest side of Stapafell (subsite 2) comprise partly isolated pillows set in hyaloclastite matrix. The amount of hyaloclastite matrix varies from 10 to 80 per cent. Subsites span a vertical distance of 20–35 m. Magnetizations were determined for 10 samples (Table 2a and b).

Site D: Lake Kleifarvatn

The rocks at Kleifarvatn form a 20 m thick sequence of compact pillowed flows overlain by stratified hyaloclastite. This sequence of lithologies suggests that the pillows formed close to the critical water/ice depth where pillows cease to form in favour of the production of hyaloclastite. Because pillow ledges are horizontal to subhorizontal we conclude that post-depositional tilting of the Kleifarvatn strata is less than 5°. The age of the Kleifarvatn strata is uncertain but these normally magnetized rocks are probably of Brunhes age. Magnetizations were determined for 12 samples (Table 2a and b) collected at one site (Table 1).

Site E: Kalfstindar

The Kalfstindar lavas are several subglacially formed volcanic ridges of olivine tholeiitic basalt (Jones 1969a, b, 1970; Fridleifsson *et al.* 1982). The sequence extends upward from compact pillowed flows to hyaloclastite beds intercalated with isolated pillows and sheet flows in a hyaloclastite matrix, to primarily hyaloclastite at the top of the sequence.

No evidence for post-depositional tilting of the Kalfstindar ridges was observed. The Kalfstindar subglacial strata are of Brunhes age, and on the basis of its young morphology, we can conclude that these ridges formed <0.1 Ma ago during the Weischelian Glacial.

Four subsites were sampled at this locality (Table 1), totalling 22 samples (Table 2a and b). Subsites cover a vertical distance of 289 m, but are up to 200 m apart horizontally.

PALAEOMAGNETIC MEASUREMENTS

From each 2.5 cm diameter core a sample of approximately 8 cm^3 was cut for palaeomagnetic measurements. These measurements were made using a Schonstedt digital spinner magnetometer at Dalhousie University. The samples were demagnetized in 12 successive steps up to 50 mT.

In almost all cases [the exceptions are noted in Table 2 (a and b)] the magnetizations were stable so the results given in Table 2 are the averages of all 12 steps. Typically the standard deviation of the inclination is about 1° while that of the declination is less than 3° when using all 12 demagnetizing steps. The greater scatter in declination is a result of the generally steep inclination.

Within-pillow variation

Two or three cores were measured from 16 pillows (Table 2a). Multiple samples from 10 individual pillows have very similar magnetic directions, with α_{95} between 1.9° and 13.2° (Table 2a). Four pillows have α_{95} around 20°, while two others have even larger values. For comparison, a study of the variation in the direction of NRM of four pillows from the Mid-Atlantic Ridge at 45°N latitude, showed within-pillow variation on as many as 20 core slices per pillow of less than 2° (Ryall & Ade-Hall 1975). Part of the greater variation in the Iceland pillows may be due to an error in orientating three cores from one pillow rather than a number of slices from a longer core. Alternatively, these variations may be recording moderate amounts of rotation of the less compact pillowed flows as different regions of the pillow cools through the Curie temperature with rare significant rotations resulting in spurious polarities. In one case (pillow B3–8), a reversely and a normally magnetized sample were collected from the same pillow, with an angular difference between the magnetic vectors of 115° and difference between the inclinations of 75.5°.

The variation of intensity of remanent magnetization (*Mo*) within pillows is usually less than 10 per cent, with only one pillow (A4-6) having a difference as much as 20 per cent (Table 2a).

Within-site variation

Palaeomagnetic results for each pillow, either averages from Table 2(a), or individual measurements, are given in Table 2(b) along with the averages for each subsite. The subsite average inclination, declination and α_{95} are plotted in Fig. 3(A–E). In order to allow direct comparison with seafloor basalts, for which declinations are not normally available, we have also calculated average inclinations along with their standard deviations. The inclinations thus calculated seldom differ by more than a degree from the average inclinations obtained using the declinations as well.

Site A

Measurements for pillows are given in Table 2(b) grouped into eight subsites. The subsite average directions with α_{95}s are plotted in Fig. 3(A) along with the present-day direction and the direction for a centred axial dipole. Only two subsites do not include the dipole direction within their circle of α_{95}. The maximum angular difference between subsite directions is 22° (Fig. 3A) but can be much greater, up to 38°, between individual pillow directions.

Apart from subsite A9, the standard deviation of the magnetization at a subsite is less than 17 per cent. The average for the site is 9.42 ± 0.99 A m^{-1}. Almost all of the magnetizations are fairly hard with Median Demagnetizing Fields (MDF's) ranging from 34 to 50 mT.

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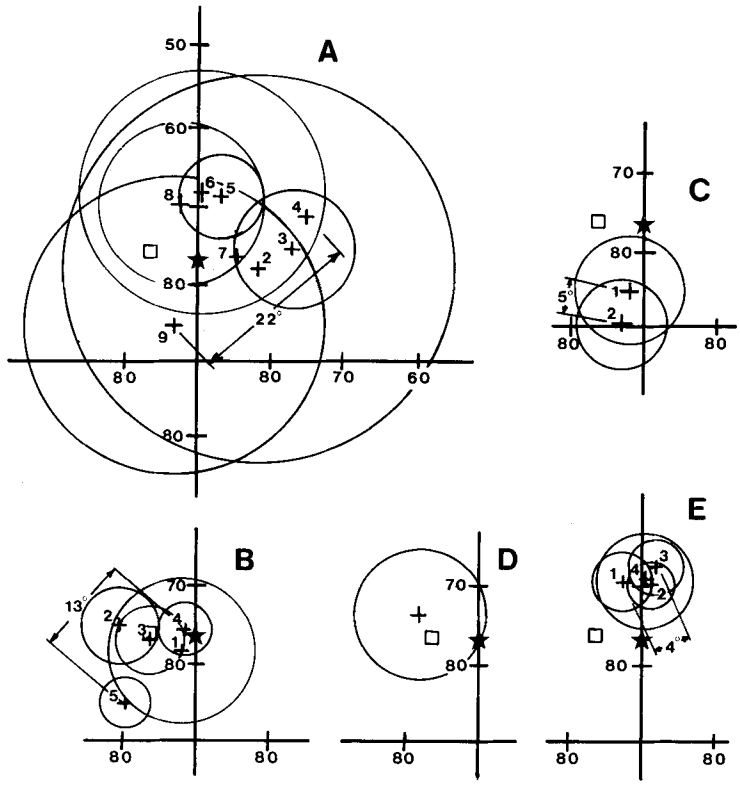


Figure 3. Wulff Stereonet plots of stable inclination and declination measurements and alpha 95 for each of the sites. The solid star is the dipole inclination. The square is the present field direction. Note: alpha 95 for site A7 is too large to plot on this figure.

Site B

Although only compact pillow flows were sampled, two of the nine have inclinations that are non-dipolar (B3-8A and B3-14). Excluding these two samples, α_{95} of the subsite average direction is less than 9°, usually less than 4°. Two of the five subsites include the dipolar direction within their circle of confidence. The maximum angular difference between subsites is 13° (Fig. 3B) but as much as 29° between individual pillows (excluding reversed samples).

Once again the subsite range of magnetization is moderate with standard deviation less than 20 per cent. The average for the site is 14.13 ± 2.48 A m^{-1}. Again most of the MDF's are fairly hard varying from 35–50 mT; however, four samples are soft (MDF's 13–15 mT). This soft magnetization is not correlated with either direction or intensity.

Site C

At site C, random tilting of up to 20° of cooling ledges suggests syndepositional rotation of pillows. However, the tight clustering of magnetic directions (α_{95} less than 7° at each subsite) indicates that the rotation of the pillows occurred before they cooled below Curie temperature. The two subsite average directions are only 5° apart and overlap each other (Fig. 3C), but do not include either the dipole direction or the present field direction. The maximum angular difference in the measured magnetic directions is 17°. Three of the 13 pillows were sampled twice (Table 2a) and although inclinations for cores A and B from an individual pillow were always very close, the declinations in two cases were very different giving an angular difference of 25° between the cores from pillow C1-23.

The standard deviation of subsite magnetizations is less than 17 per cent with a site average of 16.97 ± 0.37 A m^{-1}. MDF's at this site range from 16 to 60 mT with all subsite C1 samples having values less than 31 mT.

Site D

The 12 pillows (Table 2b) were from one subsite consisting of compact pillowed flows. The average magnetic direction when plotted, Fig. 3(D), just includes the dipolar direction. The amount of scatter, as measured by α_{95} is large (8.3°) for a subsite with so many samples. All other subsites with more than five samples have α_{95} less than 6°. Because local field conditions at site D limited geographical sighting for core orientations to distances within 5 km, whereas for other sites distances were typically 10–40 km, the larger scatter at D may be due to lower accuracy in orienting the drilled cores.

The magnetization at this subsite, 15.91 ± 4.4 m^{-1}, was similar to sites B and C, although the variation was greater with a standard deviation of 28 per cent. The magnetizations were relatively soft: all MDF's, except D1-21, are less than 22 mT.

Site E

The 22 pillows of site E were collected from four subsites. The four subsite directions are grouped closely together with relatively small values of α_{95} which nonetheless all overlap.

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However, none of these includes the dipole direction or the present field direction within its circle of confidence. Two pillows from subsite E4 which had highly anomalous directions were excluded from the averaging process. Two of the subsites (E1 and E2) have similar magnetization, 10.32 and 10.91 A m^{-1}, with low standard deviations. (<15 per cent) and relatively uniform MDF's (39–51 mT). The other two subsites show a much greater range of intensity and MDF. Yet subsite E3, which has the greatest variation of intensity and MDF, has very tightly grouped directions with $\alpha_{95} = 2.9°$.

Summary

Averages for the sites, giving equal weight to the subsite values (a) and equal weight to the pillow values (b) are given in Table 3 and plotted in Fig. 4. It is noticeable that the α_{95}s are lower for averages calculated using pillow values rather than subsite values. This is because the expression for α_{95} contains a term to the power of $1/(N-1)$ which increases dramatically for small N. The same factor contributes to the larger α_{95}s for the subsites of site A each of which have only two to five samples while most other subsites have closer to 10 samples. No matter which averaging procedure is used for a particular site, the resulting direction is essentially the same; it is only the size of the circle of confidence that differs. Under either process of averaging, the directions for site B and D are superimposed and include the present field direction within their circles of confidence. Likewise, the directions for sites C and E are clearly distinct and include neither the present field direction nor the centred axial dipole direction. Considering the site-average inclinations determined from inclinations alone, site B includes the dipole direction within its standard deviation, site D almost does and sites B and D are within $\sim 2°$. Only site E is clearly distinct. If one uses 2σ which is a more appropriate comparison to α_{95} then only site E does not include the dipole direction within 2σ. It should also be noted that the average inclinations in column 3 of Table 3 are always less than the averages of column 4. This is an inevitable consequence of geometry.

Site average magnetizations range from 9.42 A m^{-1} for site A to 16.97 A m^{-1} for site C with an overall average of 13.4 A m^{-1}. Site A is the oldest and has high vesicle content and is the only site where hydrothermal alteration had formed noticeable secondary amygdales. It is probable that the high vesicle content and secondary alteration contributes to the low magnetization and possibly the scatter of directions.

There is no obvious correlation between α_{95} and the occurrence of pillow breccias, hyaloclastite or depositional dip. For example, at site D only compact pillows were sampled but α_{95} is 8.3° whereas at subsite E3 pillow flows are intercalated with hyaloclastite yet α_{95} is only 2.0°. Likewise at subsite C2 pillow breccias with 5–10 per cent hyaloclastite were sampled and α_{95} is 5.7°. Even in the case of subsites A6 and A7 where pillow breccias intercalated with hyaloclastite with a depositional dip of 40° were sampled, α_{95} is high, 15.8° and 52.2° respectively, yet the standard deviation of the inclination is roughly 5° in both cases.

In addition, there is no obvious correlation between α_{95} and either intensity or hardness (as determined by MDF) of magnetization. Site E3 which has the widest range of intensities and MDF's has the same α_{95} (2.9°) as site B5 which has the smallest range. There is no correlation between intensity and MDF; a plot of one against the other produces a random scatter.

A summary may be given as follows.

(1) The directions of all but a handful of the cores are stable during AF demagnetization to 50 mT and are independent of the strength or hardness of the magnetization.

(2) The average directions for subsites have varying circles of confidence but α_{95} does not seem to depend on the occurrence of brecciation, the presence of hyaloclastite etc.

(3) Despite these well-behaved magnetic directions, fewer than half of the subsites include the direction of the centred axial dipole field within their α_{95}.

Causes of anomalous polarities

Two rocks sampled from site B have anomalous directions. These samples (B3-8A and B3-14) were sampled from a pillow or subsite from which other, normally magnetized samples were collected.

It is unusual to sample rocks of opposing magnetic polarity from the same pillow or flow even if the rocks have been altered (Ryall & Ade-Hall 1975)—although some dispersion of directions has been recorded (Henley, Johnson

Table 3. Site averages.

SITE	J_o (σ)	I (σ)	I^a	D^a	(α95)	I^b	D^b	(α95)
A	9.42 (0.99)	70.8 (4.0)	73.4	15.9	(6.3)	73.3	11.9	(4.3)
B	14.13 (2.48)	75.9 (2.8)	76.9	333.8	(5.3)	77.3	339.1	(2.5)
C	16.97 (0.37)	82.9 (0.0)	86.7	314.6	(9.6)	87.8	329.1	(4.6)
D	15.91 (4.40)	67.7 (7.8)	72.0	334.8	(9.2)	72.0	334.8	(9.2)
E	11.55 (2.36)	67.5 (2.0)	68.9	0.7	(2.3)	68.0	0.2	(2.1)

J_o (σ) - Intensity of NRM in units of Am^{-1} with standard deviation.
I (σ) - Inclination with standard deviation.
I, D (α95) - Inclination and declination with alpha 95.
a. Average using subsite values.
b. Average using pillow values.

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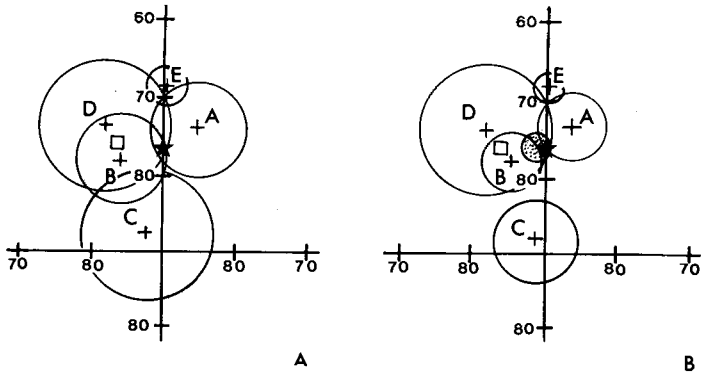


Figure 4. Wulff stereonet plot of stable directions averages for each site. (a) Site averages giving equal weight to each subsite. (b) Site averages giving equal weight to each pillow. Small shaded circle is grand average field direction.

& Lewis 1983; Soroka & Beske-Diehl 1984). However, a similar occurrence of within-flow opposing magnetic polarities within Bruhnes age rocks was recognized in two subaerial flows by Heller (1980) and Heller & Peterson (1981), and in two subaqueous flows from the Narrowgate rift valley of the Mid-Atlantic Ridge by Van Wagoner & Johnson (1983). In both of these cases the reversely magnetized samples have elevated Curie temperatures, raising the suspicion of self-reversal. In a study of the magnetic properties of Bruhnes age basalts from the Mid-Atlantic Ridge at 45 °N (Ade-Hall *et al.* 1973) a small fraction of reversely magnetized basalts with typical Curie temperatures were found intermingled with those of normal polarity.

Samples B3-8a and B3-14 with the non-dipolar and shallow inclinations have Curie temperatures of 240 °C as determined from $J_s(T)$ curves, which show no signs characteristic of self-reversing mineralogy. In addition, the declination and inclination values for samples B3-8A and B3-14 are stable throughout demagnetization to 50 mT, suggesting that they represent the initial NRM. These results suggest that secondary magnetization, or self-reversal is not a likely cause of the shallow, reversed polarities. Because the anomalous samples are from parts of pillows or units from which other normally magnetized rocks were collected, and because the reverse inclinations are shallow (i.e. non-dipolar) it is unlikely that the reversals are due to short magnetic reversals within the Bruhnes epoch.

All laboratory measurements have been carefully checked and the anomalous directions confirmed. Two possibilities remain. Either the cores were misoriented in the field or the pillows have rotated either during cooling (B3-8A) or after cooling (B3-14).

DISCUSSION AND CONCLUSIONS

The results of this study on subglacial pillows indicate the following.

(1) Differences in the magnetic directions of pillows at the same site of up to 38° (ignoring the two anomalous results).

(2) Differences in the magnetic direction of subsites of up to 22°. These differences are in the absence of any appreciable tectonic rotation of the rock units.

(3) The use of the standard deviation of inclination only (as in the case of DSDP cores) may make the directions look more tightly grouped than they are.

(4) There does not appear to be any relationship between the degree of brecciation or percentage of hyaloclastite surrounding pillows and the scatter of magnetic vectors. This indicates that, as with compact pillowed flows, pillow breccia units which erupted in shallow water have come to a rest by the time they have cooled below the Curie temperature. Therefore these units can be used for palaeomagnetic studies with as much confidence as the compact pillowed flows as long as the pillowed portions of the breccias are sampled.

Considering points (1) and (2) above and examining Figs 3 and 4 we came to the following conclusions.

(1) At any subsite there will be a scatter of magnetic directions of the pillows, perhaps due to some rotation of the individual pillows after they have acquired their magnetization.

(2) The directions for the pillows at a typical subsite will average to a direction with an α_{95} usually less than 10° (if $N > 10$) which has about a 50 per cent chance of including the centred axial dipole direction.

(3) The cause of the different site average directions is secular variation. The variation of directions is within the range reported by Wilson, Dagley and McCormack (1972) from lavas and Creer & Tucholka (1983) from lake sediments.

The grand average palaeofield direction (using pillow values) is $I = 75.5°$, $D = 354°$, $\alpha_{95} = 2.2°$ (which does include the dipole field direction) and angular standard deviation of 11.6°. The latter value is about half that reported by Kristjansson (1968) from a study in NW Iceland but is only about 1° higher than the best-fit curve of Brock (1971) so we may conclude that we are indeed measuring typical standard deviation.

This study raises doubts about the tectonic significance of small amounts of tilt (<30°) estimated from drill cores on the basis of palaeomagnetic inclinations (e.g. Hall & Robinson 1979)—probably the main cause is secular variation. We also suggest that some of the tilts determined from DSDP cores on the basis of segregation vesicle studies (Van Wagoner 1986) are the result of variable rotation as the pillows cool through the solidus temperature (~1200°–1000 °C). Our study indicates that it is not uncommon for

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pillows to rotate by as much as 30° with respect to one another after they cool below the Curie temperature (150°–300 °C). This problem of rotation of rocks as they pass through the solidus temperature was previously discussed by Van Wagoner & Johnson (1983; see also Van Wagoner 1986) and Bideau *et al.* (1984). Based on our study it seems as though both megascopic and microscopic geopetal features may only be used as an estimate of the initial cooling position of the rock as it passed through the Curie temperature within the range of 30°. Consequently, one must be very cautious in ascribing structural significance to tilt estimates of less than 30° based on magnetic inclinations in DSDP or ODP cores.

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